# Practice workbook · Explain how a mutation changes a protein and phenotype

Chaining a base change through codons and amino acids to a faulty or missing protein and a named condition or characteristic.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-MUTATION-TO-PHENOTYPE

# 1. Question 7 · part a · 3 marks

January 2023 Paper 1

Worked-example retry · Attempt part a only.

**7** People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

**(3)**

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

**(2)**

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**20**

P 7 2 2 7 1 A 0 2 0 2 4

## 1. Question 7 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 8 · part c · 4 marks

June 2021 Paper 2

Further practice · Attempt part c only.

**8** Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

**(2)**

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

**(1)**

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

**(4)**

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**20**

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## 2. Question 8 · part c · 4 marks

June 2021 Paper 2

Further practice · Attempt part c only.

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(ii) Calculate the probability that these parents will produce a boy with the condition.

**(3)**

probability = .....

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(c) White forelock is a result of a mutation in a molecule of DNA.

Describe how this mutation causes the white forelock phenotype.

**(4)**

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**(Total for Question 8 = 14 marks)**

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## 2. Question 8 · part c · 4 marks · Working and answer

Working and answer

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## 3. Question 5 · part a · 3 marks

January 2022 Paper 1

Further practice · Attempt part a only.

**5** (a) Describe the possible effects of a DNA mutation.

**(3)**

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

| **Age of mother in years** | 15–19 | 20–24 | 25–29 | 30–34 | 35–39 | 40–44 |
|---|---|---|---|---|---|---|
| **Probability of chromosome aberration** | 0.0010 | 0.0010 | 0.0010 | 0.0015 | 0.0045 | 0.0140 |

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## 3. Question 5 · part a · 3 marks · Working and answer

Working and answer

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# 4. Question 8 · part a · 4 marks

June 2022 Paper 2

Further practice · Attempt part a only.

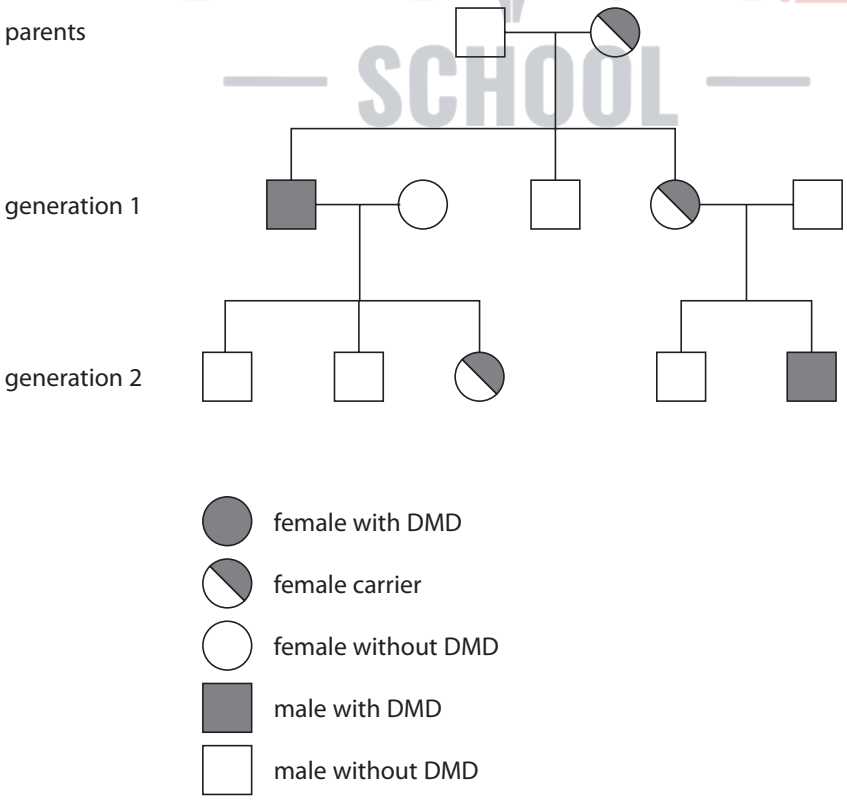
**8** (a) Duchenne muscular dystrophy (DMD) is a sex-linked genetic disorder caused by a gene mutation.

The symptoms of DMD are caused by the absence of a protein called dystrophin.

Explain how a gene mutation can result in the absence of dystrophin.

**(4)**

(b) The diagram shows the inheritance of Duchenne muscular dystrophy in one family.




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## 4. Question 8 · part a · 4 marks · Working and answer

Working and answer

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## 5. Question 6 · part a.i · 4 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

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**6** Sickle cell anaemia is an inherited condition. It is caused by a mutation of the gene that controls the production of haemoglobin.

This condition is caused by a recessive allele.

(a) (i) Explain how a mutation can cause a condition such as sickle cell anaemia.

**(4)**

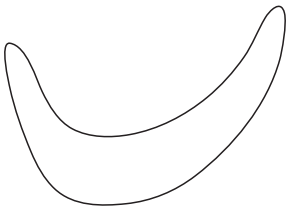
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# 5. Question 6 · part a.i · 4 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

(ii) In low oxygen concentrations the red blood cells of someone with sickle cell anaemia become like the cell shown in the diagram.



Explain why a person who has the sickle cell condition often feels tired.

**(5)**

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## 5. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

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## 6. Question 5 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

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**5** DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

**(2)**

number of codes = .....

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

**(3)**

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(ii) State **one** factor that can cause mutation.

**(1)**

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

**(2)**

## 6. Question 5 · part b.i · 3 marks · Working and answer

Working and answer

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## 7. Question 5 · part b.iv · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iv only.

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**5** DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

**(2)**

number of codes = .....

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

**(3)**

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(ii) State **one** factor that can cause mutation.

**(1)**

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

**(2)**

## 7. Question 5 · part b.iv · 1 mark

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Further practice · Attempt part b.iv only.

(iv) State why some mutations have no effect on the phenotype of an organism.

**(1)**

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**(Total for Question 5 = 9 marks)**

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Working and answer

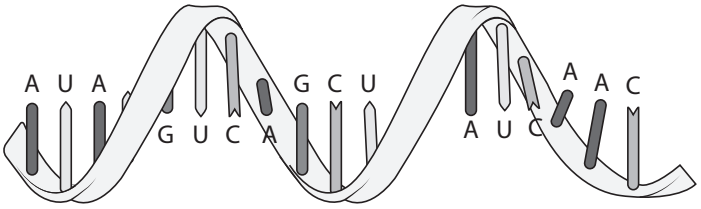
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# 8. Question 2 · part b · 3 marks

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Further practice · Attempt part b only.

**2** (a) The diagram shows a strand of RNA found in a body cell.




(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

**(2)**

1 .....

2 .....

(ii) State two differences between RNA and DNA.

**(2)**

1 .....

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2 .....

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(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

**(1)**

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# 8. Question 2 · part b · 3 marks

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Further practice · Attempt part b only.

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(b) Before a cell divides by mitosis, the DNA inside the cell is replicated.

When DNA is replicated, mutations can happen.

Explain how mutations can affect the final product made.

**(3)**

**(Total for Question 2 = 8 marks)**

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Working and answer

## 9. Question 5 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

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**5** (a) (i) Describe what is meant by the term **mutation**.

**(2)**

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

**(3)**

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*Turn over* ▶

# 9. Question 5 · part a.ii · 3 marks

November 2024 Paper 2

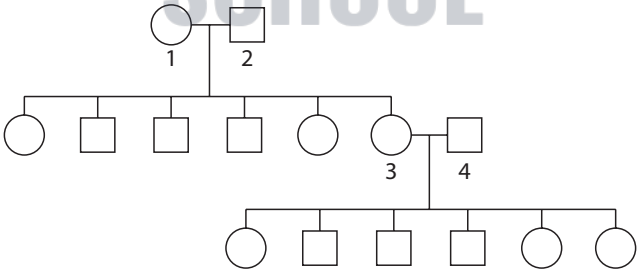
Further practice · Attempt part a.ii only.

(b) (i) The condition known as hairy ears is thought to be caused by a mutation of an allele on the Y chromosome.

Explain how the genetics of this sex-linked condition differ from the genetics of haemophilia.

**(2)**

(ii) The diagram is a family pedigree showing the inheritance of the hairy ears mutation.




Person 1 is a female without the condition and person 2 is a male with the condition.

State the genotype of person 1 and the genotype of person 2.

Use H to represent the allele for hairy ears.

**(2)**

person 1

person 2

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## 9. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

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# 10. Question 7 · part a.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.i only.

**7** Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

**(2)**

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

**(2)**

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## 10. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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